Herbert Wertheim College of Engineering
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EIN 4360 Facilities Planning and Material Handling

Automated Storage & Retrieval Systems

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POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

Summary

- Define Rack-Supported ASRS
- Discuss space considerations in ASRS
- Learn how to calculate ASRS cycle times
  - Differences between single and dual command times

Automated Storage & Retrieval Systems (ASRS)

- ASRS are used to handle and store handle and store raw materials, work-in-process inventories and finished goods

- Although an ASRS can be installed within an existing building, larger AS/R systems are often built as stand-alone structures (Rack Supported ASRS)

Monde Nissin, Philippines
Automated Storage &
Retrieval System
DEMATIC

Benefits

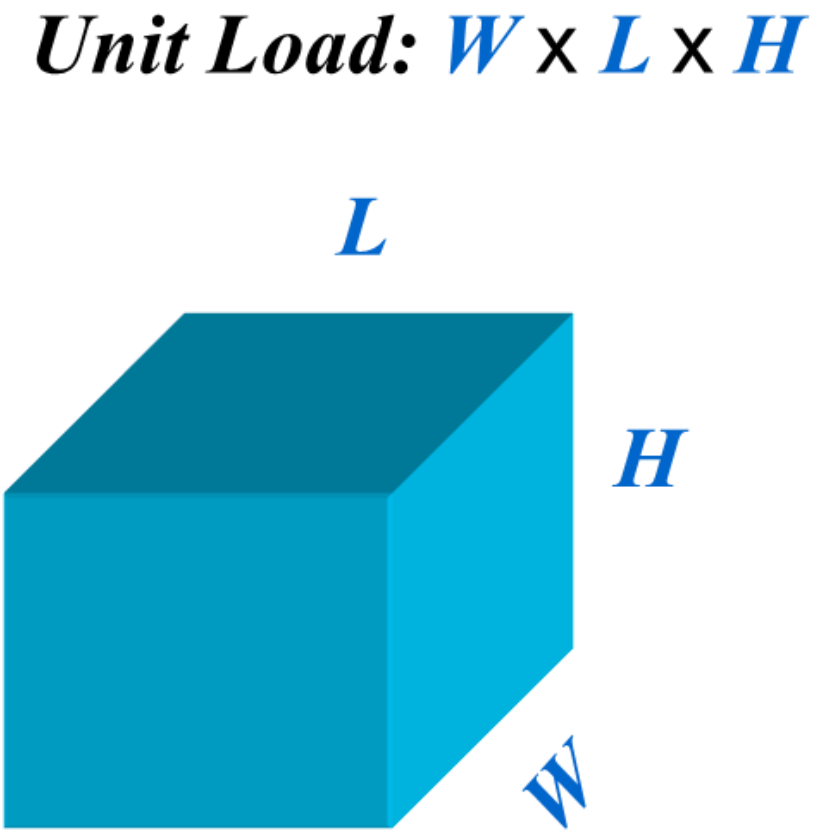
- Improved Space Utilization
- 75% of rack space can be recovered
- High saving in pickers time
- Allows quick access to goods via a system of shelves operating on a shuttle mechanism
- It brings the exact pick/store location to the operator
- Increases retrieval productivity by more than 2.5 times

Considerations

- Access area of the system is ergonomically designed to present stored items at an ideal height for picking
- High employee safety and improved throughput

Considerations

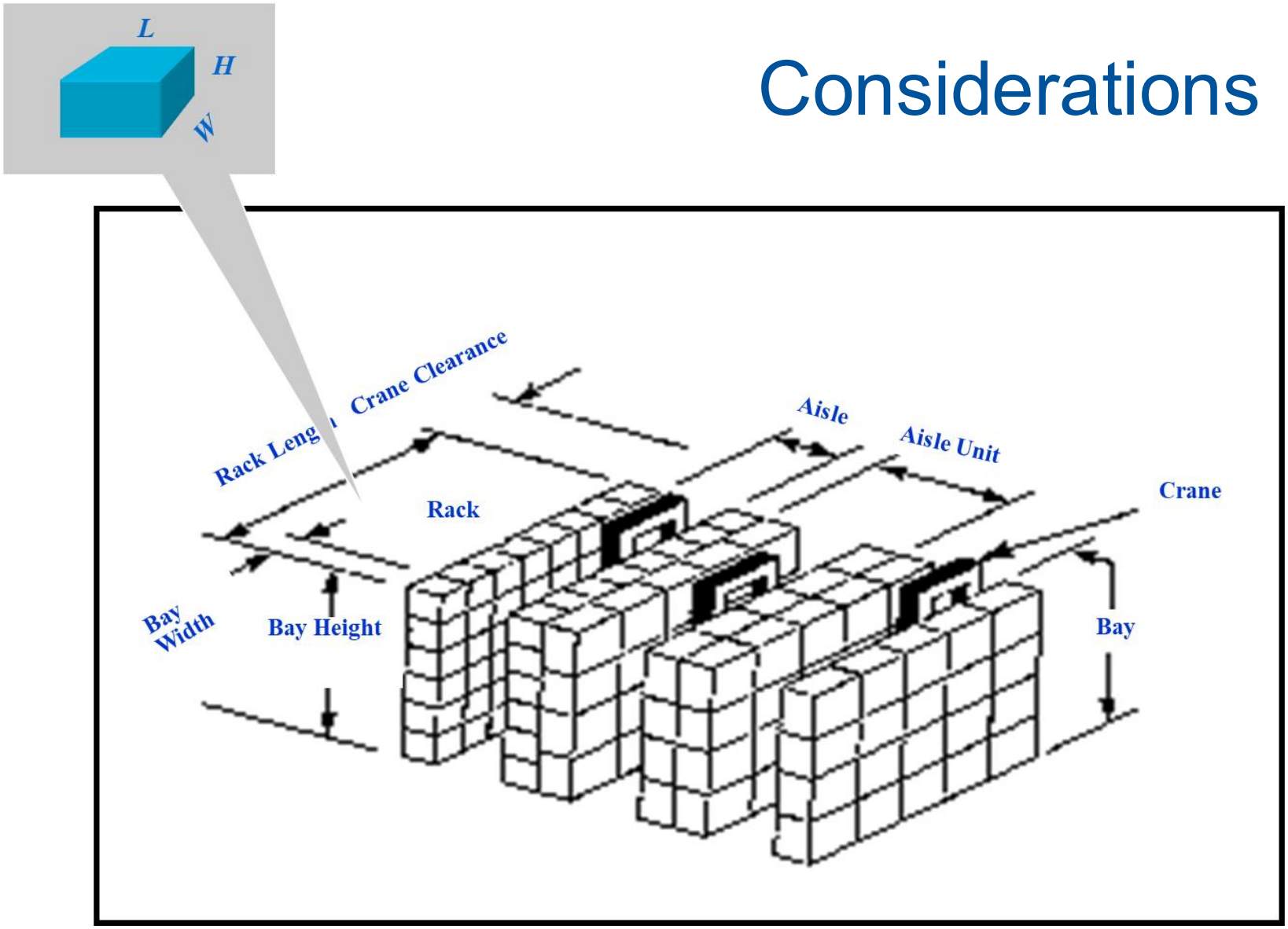
- Need to determine the volume of a unit load
- In most applications we can assume that the unit load will be rectangular
- So, the calculation is just:
  - Volume=WxLxH



Considerations

- **Bay:** A vertical stack of unit loads from floor to ceiling
- **Row (Rack):** A series of bays located side by side
- **Aisles:** Spaces between rows
- The aisles are used for stacker cranes to move up and down between rows
- Each crane serves both sides of an aisle
- Light loads of less than 2500 pounds require a 6-inch clearance for rack support and crane entry
- Heavier loads require 9 inches

Considerations



Some Facts

- Racks may be 80 - 90 feet high
- Served by computer-controlled stacker cranes carrying 3k-4k pound loads
- Can travel at speeds of 500 feet per minute in aisles that are only 6 inches wider than the cranes
- Savings of as much as 20 percent of the inventory cost
- Fully mechanized system requiring little labor to operate
- Single operator can handle a warehouse with 1000's of parts
- Minimizes need of material handling equipment and material handling operations

Design Process

1. Dimensions and weight of the load to be stored
2. Number of units to be stored
3. Throughput rate per hour
4. Number of cranes needed (single or dual cycle)
5. Number of rows required
6. Building height and load height
7. Number of bays
8. System length
9. System width

Equations

$$\textit{Number of cranes} = \frac{\text{throughput per hour}}{(\text{cycles per hour}) \times (\text{efficiency})}$$

$$\textit{Loads stacked per bay} = \frac{\text{height of storage building}}{(\text{load height} + \text{clearance between stacks})} - 1$$

or

$$\textit{Loads stacked per bay} = \frac{\text{height of storage building} - (\text{ceiling} + \text{floor clearance})}{(\text{load height} + \text{clearance between stacks})}$$

$$\textit{Bays per row} = \frac{\text{number of units to be stored}}{(\text{number of rows}) \times (\text{number of loads vertically stacked per bay})}$$

System width = (aisle unit) x (number of cranes)

Note: *aisle unit* should be at least 3 x L + 2 ft for clearance.

System length = (width of bay + clearance) x (bays per row)+ (crane clearance)

Note: crane clearance is usually 25'.

Example

A manufacturing company wishes to store a unit on a 48"x 36" x 24" (L x W x H) pallet having a weight of 1,400 lbs and 75 dual cycles per hour. The total storage is 18,000 unit loads. The height of the building is 80', but clearances of 4.5' from the ceiling and 6" from the floor for rack support are needed. A clearance equal to 6" between stacks and an availability factor of 85% for cranes are considered.

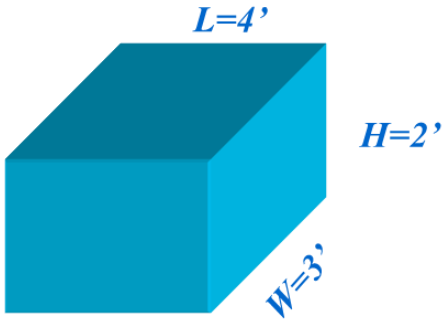
a) Determine the number of stacks (loads stacked per bay) that can be accommodated with height of the load.

b) Determine the number of dual cranes needed (assume each dual crane performs 22 cycles/hour).

c) Determine the number of bays needed.

d) Determine the storage dimensions (width, length, height)

Example Data



- W = 3', L = 4', H = 2'
- Throughput = 75 dual cycles per hour
- Dual crane cycles/hour = 22
- Total storage unit loads = 18,000
- Building height = 80'
- Clearances:
  - 4.5' from the ceiling
  - 0.5' from the floor for rack support
  - 0.5' between stacks
- Dual crane availability factor = 0.85

Example Solution

a) Loads stacked per bay = $\frac{80'}{2' + 0.5'} - 1 = 31$

b) Number of dual cranes = $\frac{75}{22 * 85\%} = 4.01$ Rounded up to **5**

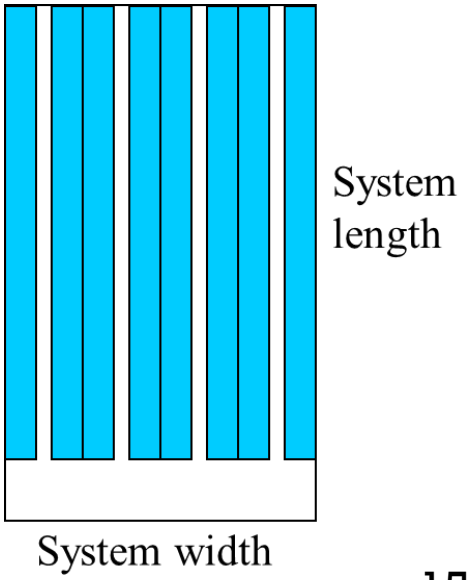
c) Number of rows = $2 * 5 = 10$

Bays per row = $\frac{18,000}{10 * 31} = 58.06$ Rounded up to **59**

d) Width = (aisle unit) x (cranes) = $(3 * 4' + 2') * 5 = 70'$

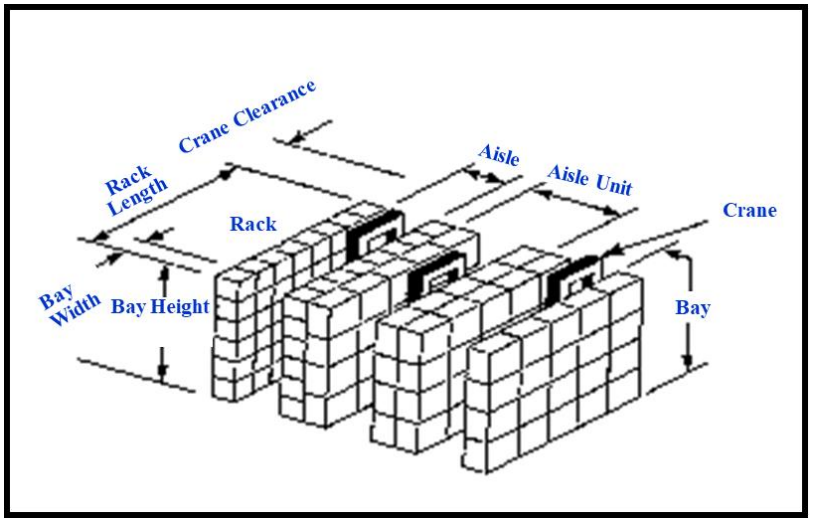
Length = $(3' + 0.5') * 59 + 25' = 231.5'$

Height = $80'$



Considerations

- **Remember:** Typically, each S/R machine operates in a single aisle and services storage racks on either side of the aisle.



Machine Cycle Times

- **Time required** to perform an operation (store/retrieve)
  - Expected S/R machine cycle time formulas will be used
  - S/R machine can travel horizontally and vertically simultaneously
  - Time for travel (pickup/deposit (P/D) station, storage/retrieval (S/R) location)
    - =max {horizontal, vertical} travel times -- Chebyshev travel
- **Single** command – either storage or retrieval operation
- **Dual** command cycle – both storage and retrieval operations
- **Assumption:** cycle starts at P/D station, and ends at P/D station

Notation

- E(SC) = expected travel time for a single command cycle
- E(DC) = expected travel time for a dual command cycle
- E(TB) = expected travel time from storage location to the retrieval location during a dual command cycle
- L = rack length (ft)
- H = rack height (ft)
- h_v = horizontal velocity of S/R machine (ft/min)
- v_v = vertical velocity of S/R machine (ft/min)
- $t_h = L/h_v$ = time required to travel horizontally from P/D station to the farthest location in the aisle
- $t_v = H/v_v$ = time required to travel vertically from P/D station to the farthest location in the aisle

Expected Travel Times

- **"Normalization" of the Rack**
  - Dividing the rack's shorter (in time) side by its longer (in time) side
  - "Scaling factor" $T = \max(t_h, t_v)$ – longer (in time) side of rack
  - "Shape factor" $Q = \min(t_h/T, t_v/T)$ – ratio of the shorter (in time) side to the longer (in time) side of rack
  - Normalized rack is Q units long in one direction and one unit long in the other direction ($0 < Q \leq 1$)
  - The rack is **"square-in-time"** (SIT) if $Q = 1$
- **Assumptions**
  - Randomized storage
  - End-of-aisle P/D station located at lower left-hand corner of rack
  - Constant horizontal and vertical S/R machine velocities
  - Continuous approximation of storage rack

Expected Travel Times

(Bozer and White, 1984)

"Scaling factor" $T = \max(t_v, t_h)$
"Shape factor" $Q = \min(t_h/T, t_v/T)$

- $E(SC) = T \times [1 + Q^2/3]$
  - Single command
- $E(TB) = [10 + 5Q^2 - Q^3] \times T/30$
  - Storage location to retrieval location
- $E(DC) = E(SC) + E(TB) = [40 + 15Q^2 - Q^3] \times T/30$
  - Dual command
- E(SC) and E(DC) are minimized at Q=1.
  - (i.e. given a rack with fixed area but variable dimensions).
- Time required to pick-up or deposit load: $T_{P/D}$
  - Expected single command cycle time $T_{SC} = E(SC) + 2T_{P/D}$
  - Expected dual command cycle time $T_{DC} = E(DC) + 4T_{P/D}$

In-class Activity

Consider a rack system with a storage rack of dimensions (inches) of 56m (length) by 58n (height), where m is the number of horizontal addresses and n is the number of vertical addresses. Assume m=80, n=12, and that the S/R requires 0.35 min to perform a P/D operation, travels horizontally at an average speed of 350 fpm, and travels vertically at an average speed of 60 fpm.

Determine the cycle times for single and dual command operations.

In-class Activity

Hint

$T_{P/D}$ =

h_v =

v_v =

L =

H =

$t_h = L/h_v$ =

$t_v = H/v_v$ =

T =

Q =

T_{SC} =

T_{DC} =

Average cycle time per operation is _____ min for SC and _____ min for DC

In-class Activity

Answer

$T_{P/D}$ = 0.35 min

h_v = 350 fpm

v_v = 60 fpm

L = 56" x 80 = 373.3'

H = 58" x 12 = 58'

$t_h = L/h_v$ = 1.07 min

$t_v = H/v_v$ = 0.97 min

T = max(1.07, 0.97) = 1.07

Q = 0.91

$T_{SC} = 1.07x(1+0.91^2/3) + 2x0.35 = 2.07$ min

$T_{DC} = 1.07x(40+15x0.91^2-0.91^3)/30 + 4x0.35 = 3.24$ min

Average cycle time per operation is _2.07_ min for SC and _3.24_ min for DC